Roles & Required Skills

Minimum and core skills for each target role.

Data Scientist

Summary: Builds statistical and machine-learning models to turn data into predictions and business insight.

Minimum required skills: Python, SQL, Statistics, Machine Learning, Pandas, Data Visualization

Core skills: Statistics, Machine Learning, Data Wrangling, Feature Engineering, Model Evaluation, Data Visualization, Experiment Design

Tools: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Jupyter, Matplotlib, Seaborn

Programming languages: Python, R, SQL

Fundamentals: Linear Algebra, Probability, Calculus, Hypothesis Testing, Data Structures

Data Engineer

Summary: Designs and maintains data pipelines, warehouses and infrastructure that make data usable at scale.

Minimum required skills: Python, SQL, ETL, Data Warehousing, Spark, Cloud (AWS/GCP/Azure)

Core skills: Data Pipelines, ETL, Data Modeling, Data Warehousing, Batch & Streaming Processing, Data Quality, Orchestration

Tools: Apache Spark, Airflow, Kafka, Hadoop, dbt, Snowflake, AWS/GCP/Azure, Docker

Programming languages: Python, SQL, Scala, Java

Fundamentals: Data Structures, Distributed Systems, Database Internals, Networking Basics

Data Analyst

Summary: Analyzes data and builds reports/dashboards to support day-to-day business decisions.

Minimum required skills: SQL, Excel, Power BI, Statistics, Data Visualization

Core skills: Data Cleaning, Exploratory Data Analysis, Dashboarding, Reporting, Business Acumen, Descriptive Statistics

Tools: Excel, Power BI, Tableau, Pandas, Google Sheets, Looker

Programming languages: SQL, Python

Fundamentals: Descriptive Statistics, Data Modeling Basics, Business Metrics

ML Engineer

Summary: Productionizes machine-learning models - training, serving, scaling and monitoring them in real systems.

Minimum required skills: Python, Machine Learning, Deep Learning, MLOps, Docker, API Development

Core skills: Machine Learning, Deep Learning, MLOps, Model Deployment, Model Monitoring, API Development, Pipeline Automation

Tools: TensorFlow, PyTorch, Scikit-learn, Docker, Kubernetes, MLflow, FastAPI, AWS/GCP SageMaker/Vertex

Programming languages: Python, SQL, C++

Fundamentals: Linear Algebra, Probability, Algorithms, System Design, Data Structures

Software Engineer

Summary: Designs, builds and maintains software applications and backend services.

Minimum required skills: Data Structures, Algorithms, OOP, SQL, Git, REST APIs

Core skills: Data Structures & Algorithms, Object-Oriented Programming, System Design, Version Control, Testing, Debugging, REST APIs

Tools: Git, Docker, REST/GraphQL, SQL/NoSQL Databases, CI/CD, Linux

Programming languages: Java, Python, C++, JavaScript

Fundamentals: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, OOP